

Computer Networks Module 1 Assignment

How the Internet Delivers a Web Page

Example Website: YouTube (www.youtube.com)

Submitted By : Gokuldev P S

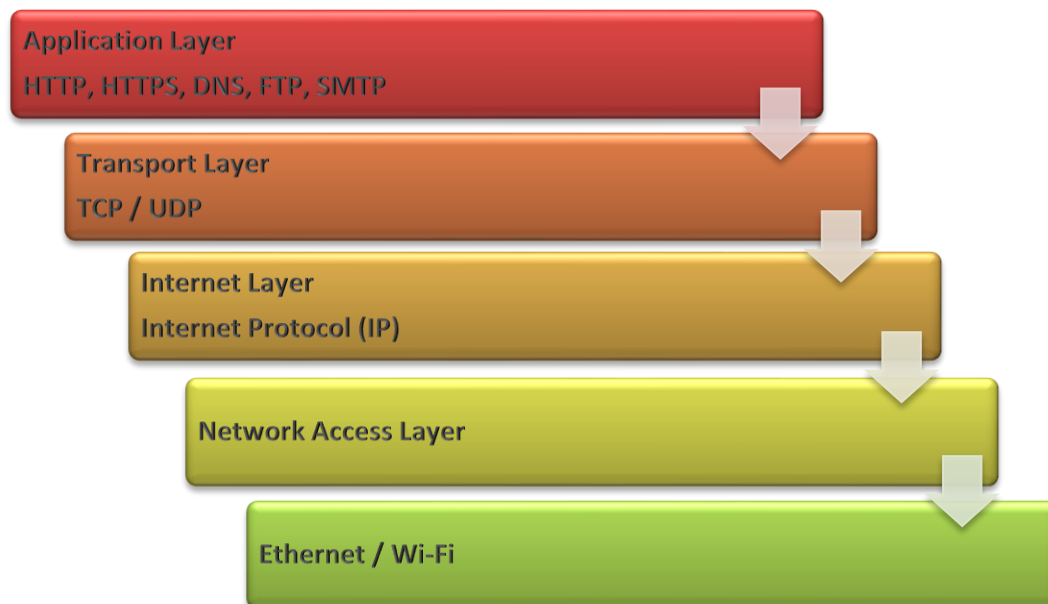
Class : S5 – CSE - A

Roll no: 31

VJC24CS093

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

TCP/IP Protocol Stack



Explanation

Application Layer

The Application Layer provides network services directly to users and applications. It enables communication between web browsers and web servers. Common protocols include HTTP, HTTPS, DNS, FTP, SMTP, and POP3.

Transport Layer

The Transport Layer provides end-to-end communication between devices. It uses TCP for reliable communication and UDP for fast communication without reliability. TCP ensures data reaches the destination without errors.

Internet Layer

The Internet Layer is responsible for routing packets across different networks. It uses the Internet Protocol (IP) to identify devices using IP addresses and determines the best path for data transmission.

Network Access Layer

The Network Access Layer manages the physical transmission of data through network hardware such as Ethernet cables, Wi-Fi, fiber optics, or mobile networks. It converts data into electrical or wireless signals.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

DNS (Domain Name System) acts like the Internet's phone book. Humans remember website names such as www.youtube.com, but computers communicate using IP addresses.

When a user enters www.youtube.com into the browser, the browser first checks its local DNS cache. If the IP address is not available, the request is sent to a DNS resolver. The resolver contacts DNS servers to find the correct IP address associated with the domain name. After the IP address is returned, the browser connects to YouTube's web server using that IP address.

Without DNS, users would have to remember numerical IP addresses instead of simple website names.

3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (HyperText Transfer Protocol) is the protocol used for communication between a web browser and a web server.

After obtaining the server's IP address, the browser establishes a TCP connection with the server. The browser then sends an HTTP request, usually a **GET request**, asking for the webpage.

Example:

```
GET / HTTP/1.1
Host: www.youtube.com
```

The server processes the request and returns an HTTP response.

Example:

```
HTTP/1.1 200 OK
Content-Type: text/html
```

The browser downloads the HTML document along with CSS, JavaScript, images, videos, and other resources. Finally, the browser renders the webpage so that it can be viewed by the user.

4. Explain How TCP Provides Reliable Communication for HTTP

TCP (Transmission Control Protocol) ensures that HTTP communication is reliable.

It establishes a connection using a **three-way handshake** before any data is transmitted. Each data packet is assigned a sequence number so that packets arriving out of order can be rearranged correctly.

The receiver sends acknowledgements (ACKs) to confirm successful delivery. If any packet is lost or corrupted, TCP automatically retransmits it. Flow control prevents the sender from overwhelming the receiver, while error detection ensures data integrity.

These features guarantee that webpages are delivered completely and correctly.

5. Identify Whether the Application Follows a Client–Server or Peer-to-Peer Architecture and Justify Your Answer

YouTube follows the **Client–Server Architecture**.

In this architecture, the user's web browser acts as the **client**, while YouTube's servers act as the **server**.

The client sends requests for webpages or videos. The server processes these requests, retrieves the required content, and sends it back to the client.

Users do not communicate directly with one another to obtain YouTube videos. Instead, all content is stored and delivered by YouTube's centralized servers.

Therefore, YouTube is a Client–Server application.

6. Compare HTTP and FTP with at Least Four Differences

HTTP	FTP
HTTP stands for HyperText Transfer Protocol	FTP stands for File Transfer Protocol
Used for accessing web pages	Used for transferring files between computers.
Default ports are 80 (HTTP) and 443 (HTTPS).	Default ports are 20 and 21.
Stateless protocol	Session-oriented protocol.
Mainly transfers HTML pages, images, CSS, and JavaScript	Transfers files such as documents, software, videos, and backups.

7. Explain How Cookies Improve User Experience with One Real-World Example

Cookies are small text files stored by a web browser on the user's computer. They save information about the user's browsing session and preferences.

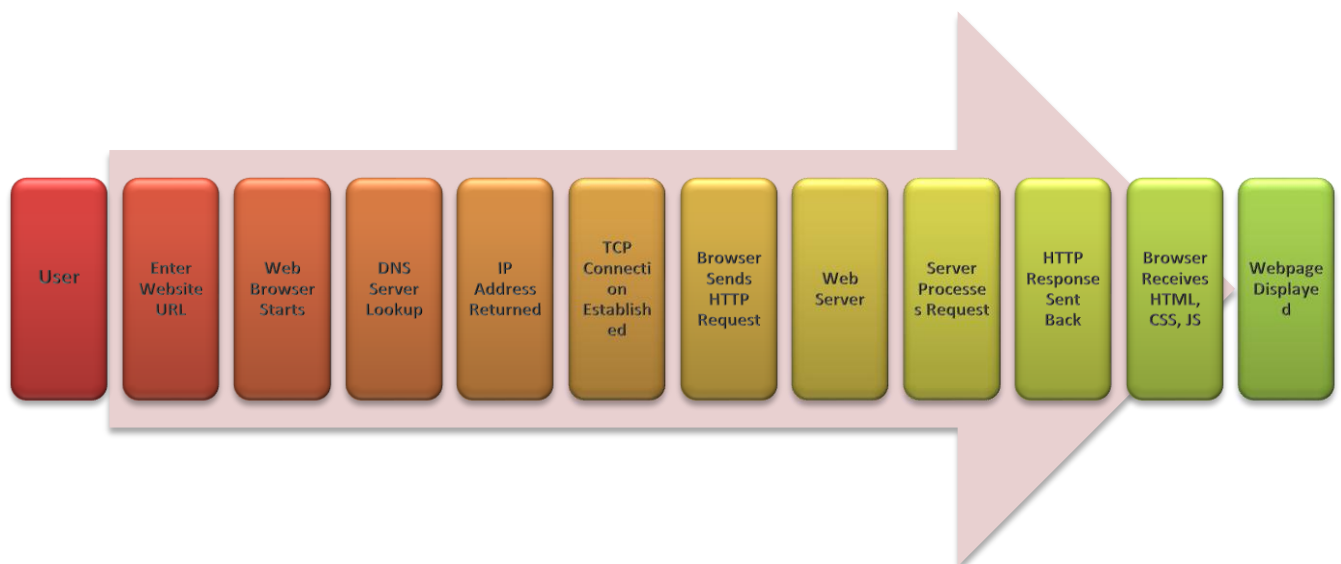
Cookies improve user experience by:

- Remembering login information.
- Saving language preferences.
- Keeping items in shopping carts.
- Providing personalized recommendations.

Real-World Example

When a user logs into **YouTube**, a cookie stores the session information. The next time the user visits YouTube, they usually remain logged in without entering their username and password again. Cookies also help YouTube recommend videos based on previous viewing history, making the browsing experience more personalized and convenient.

8. Draw a Flow Diagram Showing the Complete Communication from the Browser to the Web Server and Back



9. Write a Brief Conclusion (100–150 Words)

The Internet uses several application-layer protocols to deliver webpages quickly and reliably. DNS translates user-friendly domain names into IP addresses so that browsers can locate web servers. HTTP enables browsers and servers to exchange webpages and other web resources, while TCP guarantees reliable communication through acknowledgements, retransmissions, and error checking. Cookies further enhance the browsing experience by remembering user preferences, login sessions, and personalized settings. Together, these protocols allow users to browse websites, watch videos, send emails, shop online, and use cloud services efficiently. Understanding how these protocols work together provides a strong foundation for studying computer networks and highlights the importance of networking technologies in modern digital communication.

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